

Final Revision Cram Sheet — NMDC Steel ET (CS&IT)

Last-day recall only. Use with your CN Cheat Sheet and NMDC Awareness Sheet. Do not learn anything new from this — only confirm what you already know.

1. DBMS

Topic	Must-recall
Keys	Super key $\supset$ Candidate key (minimal super key) $\supset$ Primary key (one chosen CK). Alternate = unchosen CKs. Foreign key refs another relation's PK. PK: no NULL, unique. Unique key: allows one NULL.
Normal forms	1NF atomic values. 2NF 1NF + no partial dependency. 3NF 2NF + no transitive dependency. BCNF every determinant is a candidate key. 4NF no multivalued dep.
Decomposition	Lossless join: R1 $\cap$ R2 must contain a key of R1 or R2. BCNF is always lossless but may not preserve dependencies; 3NF can always be both.
ACID	Atomicity (all or nothing), Consistency , Isolation , Durability . Atomicity $\rightarrow$ transaction manager/recovery; Durability $\rightarrow$ logs.
Schedules	Serial $\rightarrow$ conflict serializable $\rightarrow$ view serializable. Conflict test = precedence graph acyclic . Recoverable $\supset$ Cascadeless $\supset$ Strict $\supset$ Serial (nesting direction!).
Locking	2PL = growing + shrinking phase; guarantees serializability but not deadlock-free. Strict 2PL holds X-locks till commit. Rigorous holds all locks till commit. Conservative = deadlock-free (acquires all upfront).
Indexing	Dense = entry per record; Sparse = entry per block. Primary index on ordered key field; only one clustering index per table. B+ tree: all data in leaves, leaves linked $\rightarrow$ better for range queries ; B tree stores data in internal nodes too.
SQL	DDL create/alter/drop/truncate · DML select/insert/update/delete · DCL grant/revoke · TCL commit/rollback/savepoint. DELETE = DML, row-wise, rollback possible; TRUNCATE = DDL, no rollback, faster; DROP removes structure. WHERE filters rows before grouping, HAVING after. COUNT(*) counts NULLs, COUNT(col) does not.
ER	Weak entity = no key of its own, double rectangle, identifying relationship (double diamond), total participation. Generalisation = bottom-up, Specialisation = top-down, Aggregation = relationship treated as entity.

2. Operating Systems

Topic	Must-recall
Scheduling	FCFS = convoy effect. SJF gives minimum average waiting time (optimal) but risks starvation. SRTF = preemptive SJF. RR = time quantum, fair, good response time. Priority $\rightarrow$ starvation, fixed by aging . Turnaround = Completion – Arrival; Waiting = Turnaround – Burst.
Process vs thread	Threads share code, data, files; have own stack, registers, PC. Thread switching cheaper. Process = PCB.
Sync	Semaphore : counting & binary; wait(P)/signal(V). Mutex = lock, only owner releases. Critical section needs mutual exclusion + progress + bounded waiting. Peterson's solution (2-process software).
Deadlock	4 necessary conditions: mutual exclusion , hold and wait , no preemption , circular wait . Prevention breaks one; avoidance = Banker's algorithm (safe sequence); detection = wait-for graph; recovery = kill/rollback. Starvation $\neq$ deadlock.
Memory	Paging $\rightarrow$ internal fragmentation, invisible to user, fixed-size frames. Segmentation $\rightarrow$ external fragmentation, variable size, logical view. Logical $\rightarrow$ physical via MMU. TLB caches page table entries. Effective access time = $h \cdot (\text{TLB} + \text{Mem}) + (1-h)$ (TLB+2Mem).
Page replacement	Optimal = lowest faults (not implementable). FIFO suffers Belady's anomaly (more frames $\rightarrow$ more faults); LRU and Optimal do not. Thrashing = excessive paging; fixed by working-set model.
Disk	FCFS, SSTF (starvation), SCAN/elevator, C-SCAN, LOOK, C-LOOK. Seek time dominates.
Linux	chmod numeric $r=4$ $w=2$ $x=1 \rightarrow 755 = \text{rwxr-xr-x}$, $644 = \text{rw-r--r--}$. inode stores metadata, NOT the filename . Hard link = same inode, no cross-filesystem, survives original deletion; soft link (ln -s) = new inode, points to path, breaks if target deleted. Kernel $\neq$ shell. / $\rightarrow$ bin, etc (config), var (logs), proc (process info), dev (devices), home.

3. DSA — the complexity table

Algorithm	Best	Average	Worst	Space	Stable	In-place
Bubble	n	n^2	n^2	1	Yes	Yes
Insertion	n	n^2	n^2	1	Yes	Yes
Selection	n^2	n^2	n^2	1	No	Yes
Merge	$n \log n$	$n \log n$	$n \log n$	n	Yes	No
Quick	$n \log n$	$n \log n$	n^2	$\log n$	No	Yes
Heap	$n \log n$	$n \log n$	$n \log n$	1	No	Yes
Counting / Radix	—	linear	linear	$n+k$	Yes	No

- Search: linear $O(n)$; **binary search $O(\log n)$** needs sorted array; BST search $O(h)$ — $O(\log n)$ balanced, $O(n)$ skewed.
- Master theorem:** $T(n) = aT(n/b) + f(n)$, compare $f(n)$ with $n^{\log_b a}$. $T(n) = 2T(n/2) + n \rightarrow n \log n$.
- Binary tree: max nodes at level $i = 2^i$; max nodes of height $h = 2^{h+1} - 1$; n nodes $\rightarrow$ min height $\lceil \log_2 n \rceil$. Leaf nodes = degree-2 nodes + 1.
- Traversals: inorder of BST = **sorted**. Tree reconstruction needs inorder + (preorder or postorder). Preorder+postorder alone is insufficient.
- Heap in array: parent i , children $2i$ & $2i+1$. Build-heap $O(n)$; insert/delete $O(\log n)$. Min-heap root = minimum.
- Graph: adjacency matrix $O(V^2)$ space, edge check $O(1)$; adjacency list $O(V+E)$ space. BFS $\rightarrow$ queue, shortest path in unweighted. DFS $\rightarrow$ stack, cycle detection, topological sort.
- Dijkstra $E \log V$ (no negative edges) · Bellman-Ford VE (negative OK) · Floyd-Warshall V^3 (all pairs) · Prim/Kruskal MST greedy.
- Paradigms: Greedy — fractional knapsack, Huffman, Prim, Kruskal, Dijkstra. DP — 0/1 knapsack, LCS, matrix chain, Floyd, Bellman-Ford. D&C — merge sort, quick sort, binary search.
- Stack $\rightarrow$ recursion, postfix eval, parentheses matching, undo. Queue $\rightarrow$ BFS, scheduling, buffering.

4. C Programming & OOPs

Storage classes

Class	Scope	Lifetime	Default value	Storage
auto	Block	Block	Garbage	Stack
register	Block	Block	Garbage	Register (or stack)
static	Block / file	Whole program	0	Data segment
extern	Global (all files)	Whole program	0	Data segment

- **Union size = size of largest member**; struct size = sum + padding. Only one union member is valid at a time.
- $a[i] \equiv *(a+i) \equiv *(i+a) \equiv i[a]$. Array name = pointer to first element (not modifiable). 2D arrays stored **row-major** in C.
- **strlen** excludes '\0'; **sizeof** of a string literal includes it. "abc" → strlen 3, sizeof 4.
- malloc(n) → uninitialised garbage, single argument; **calloc(n,size)** → zero-initialised, two arguments; realloc resizes; free releases. All return void*.
- Call by value = copy (default in C); call by reference via pointers. Arrays are always passed as pointers.
- Macro = preprocessor text substitution, no type checking, faster, no stack frame; function = compiled, type-checked.
- Dangling pointer = points to freed memory. Wild = uninitialised. Memory leak = allocated but never freed. Stack: local/automatic; Heap: dynamic.

OOPs — overloading vs overriding (most asked)

Basis	Overloading	Overriding
Definition	Same name, different parameters , same class	Same name and same signature , parent & child class
Binding	Compile-time / static / early	Run-time / dynamic / late
Polymorphism	Compile-time polymorphism	Run-time polymorphism
Inheritance	Not required	Required
Return type	Cannot overload on return type alone	Must match (or be covariant)

- Four pillars: **Encapsulation** (data + methods bundled, data hiding), **Inheritance** (reuse, IS-A), **Polymorphism** (many forms), **Abstraction** (hide implementation, show interface).
- Constructor: same name as class, no return type, called automatically. Types: default, parameterised, copy. Destructor: **~ClassName**, no args, no return, reverse order. In inheritance, base constructor runs **first**, base destructor **last**.
- **Diamond problem** → ambiguity from multiple inheritance; C++ solves via virtual inheritance, **Java forbids multiple class inheritance** and uses interfaces.
- Abstract class: can have both abstract and concrete methods, constructors, instance variables; single inheritance. Interface: only abstract methods (pre-Java 8), no constructor, multiple implementation allowed.
- Access specifiers (widest→narrowest): public > protected > default > private. Private members are **not** inherited/accessible in child.
- Java: **JDK** → **JRE** → **JVM**. Platform independence via bytecode. Checked exceptions = compile-time (IOException); unchecked = runtime (NullPointerException, ArithmeticException, ArrayIndexOutOfBoundsException). Finally always executes. == compares references, **.equals()** compares content. String is immutable; StringBuffer is synchronised, StringBuilder is not.
- Virtual function → run-time polymorphism via vtable. Pure virtual (=0) makes the class abstract.

5. Software Engineering

- **Model selection**: fixed clear requirements → Waterfall · unclear requirements → Prototype · high risk / large project → **Spiral** (risk analysis is its distinguishing feature) · changing requirements → Agile · fast delivery with reusable components → RAD · testing at each stage → V-model.
- **Cohesion HIGH is good; Coupling LOW is good**. Cohesion best→worst: **Functional** > Sequential > Communicational > Procedural > Temporal > Logical > **Coincidental**. Coupling best→worst: **Data** > Stamp > Control > External > Common > **Content**.
- **Cyclomatic complexity $V(G) = E - N + 2$** = number of predicates + 1 = independent paths. White-box metric.
- Black-box = functional, no internal knowledge (equivalence partitioning, boundary value analysis). White-box = structural (statement, branch, path coverage).
- **Top-down integration uses STUBS; bottom-up uses DRIVERS**. Alpha = developer site, Beta = customer site. Regression = after changes. Smoke = build sanity.
- **Verification** = "are we building it right" (static, reviews). **Validation** = "are we building the right product" (dynamic, testing).
- **COCOMO** modes: Organic (small, simple) < Semi-detached < Embedded (complex, tight constraints). Effort = $a(KLOC)^b$.
- **CMM levels 1-5**: Initial, Repeatable, Defined, Managed, Optimising. Level 1 has no defined process.
- SRS characteristics: complete, consistent, unambiguous, verifiable, modifiable, traceable. Functional = what it does; non-functional = performance, security, usability.
- UML: Use case (user interaction) · Class (static structure) · Sequence (time-ordered messages) · Activity (workflow) · State (states & transitions).
- PERT = probabilistic, event-oriented; CPM = deterministic, activity-oriented. Critical path = longest path, **zero slack**.
- Maintenance: Corrective (fix bugs), Adaptive (environment change), Perfective (enhance), Preventive (future-proof).

6. COA

- **Cache average access time** — Simultaneous: $T = h \cdot T_c + (1-h) \cdot T_m$. Hierarchical: $T = h \cdot T_c + (1-h) (T_c + T_m)$. Read the question for which!
- Mapping: **Direct** = simplest, no replacement policy needed, most conflict misses. **Fully associative** = most flexible, costliest hardware. **Set associative** = compromise, most used.
- Miss types: compulsory (first access), capacity (cache too small), conflict (mapping clash).
- **SRAM** faster, no refresh, uses flip-flops, costlier, used for cache. **DRAM** uses capacitors, needs refreshing, denser, cheaper, main memory.
- **Pipeline speedup = $n \cdot k / (k + n - 1)$** ; ideal → k. Hazards: structural, data (RAW most common), control. Fixes: stalling, **operand forwarding**, branch prediction.
- Registers: **PC** next instruction address · **IR** current instruction · **MAR** address to access · **MDR/MBR** data being transferred · **AC** accumulator · **SP** stack top.
- Addressing modes: immediate (operand in instruction) · direct · indirect · register · register indirect · **indexed** → **arrays** · **relative** → **position-independent code / branches** · auto-increment → stacks/strings.
- **DMA** transfers data without CPU involvement, uses cycle stealing — fastest I/O. Programmed I/O wastes CPU (busy-waiting); interrupt-driven is in between.
- **RISC**: fewer/simple instructions, fixed length, more registers, pipelining-friendly, hardwired control, compiler does the work. **CISC**: many complex instructions, variable length, microprogrammed.
- Hardwired control = faster, inflexible. Microprogrammed = flexible, easy to modify, slower.
- **Universal gates: NAND & NOR only**. XOR from 4 NANDs / 5 NORs. Half adder: $SUM = A \oplus B$, $CARRY = A \cdot B$.
- 2's complement range for n bits: -2^{n-1} to $2^{n-1}-1$. IEEE 754 single = 1+8+23 (32 bit); double = 1+11+52 (64 bit).

- Mod-N counter needs $\lceil \log_2 N \rceil$ flip-flops. MUX with n select lines $\rightarrow 2^n$ inputs. JK race-around fixed by master-slave.

7. Cloud & Cyber Security

- **IaaS** (infrastructure — VMs, storage; most control) · **PaaS** (platform for developers) · **SaaS** (ready software — Gmail; least control).
- Deployment: Public, Private, Hybrid, Community. Essential characteristics: on-demand self-service, broad network access, resource pooling, **rapid elasticity**, measured service (pay-per-use).
- **Scalability** = planned capacity growth; **Elasticity** = automatic dynamic scaling. **Type 1 hypervisor** = bare metal (ESXi, Hyper-V); Type 2 = hosted (VirtualBox, VMware Workstation). Containers share the host OS kernel → lighter and faster than VMs.
- **CIA triad**: Confidentiality, Integrity, Availability. Plus authentication, authorisation, non-repudiation (provided by **digital signature**).
- **Symmetric** = one shared key, fast, key distribution problem (DES 56-bit, 3DES, AES 128/192/256, Blowfish). **Asymmetric** = key pair, slow (RSA, ECC, Diffie-Hellman for key exchange).
- Encrypt with receiver's **public** key for confidentiality; sign with sender's **private** key for authentication.
- Hashing = one-way, fixed length, for integrity: MD5 (128-bit), SHA-1 (160-bit), SHA-256. Not reversible, not encryption.
- Attacks: **DoS/DDoS** (availability), MITM, phishing (social engineering), **SQL injection** (input validation flaw), XSS, ransomware, zero-day, brute force, IP spoofing, replay.
- Firewall filters traffic by rules; **IDS detects**, **IPS detects and blocks**. VPN = encrypted tunnel over a public network. SSL/TLS → HTTPS on port 443.
- Malware: virus (needs host + user action), worm (self-replicating, spreads alone), trojan (disguised), spyware, rootkit, keylogger, botnet.

8. Aptitude — Formula Recall

Topic	Formulas
Boats & streams	Downstream $D=b+s$ · Upstream $U=b-s$ · $b=(D+U)/2$ · $s=(D-U)/2$ · round-trip avg speed = $2DU/(D+U)$
Time-speed-distance	$S=D/T$ · avg speed for equal distances = $2xy/(x+y)$ · trains crossing: relative speed = sum (opposite) or difference (same direction) · 1 km/h = 5/18 m/s
Time & work	1 day work = $1/n$ · A+B together = $xy/(x+y)$ · $M_1D_1H_1/W_1 = M_2D_2H_2/W_2$ · efficiency $\propto 1/\text{time}$
Pipes & cisterns	Same as time & work; outlet pipe is negative. Fill & empty together = $xy/(y-x)$
Percentage	Increase then decrease by $x\% \rightarrow \text{net } -x^2/100\%$ · successive $a\%, b\% \rightarrow a+b+ab/100$
Profit & loss	$P\% = (SP-CP)/CP \times 100$ · $SP = CP(100+P)/100$ · two articles at same price with $+x\%$ and $-x\% \rightarrow \text{always loss of } x^2/100\%$
SI & CI	SI = $PRT/100$ · CI: $A = P(1+R/100)^n$ · CI-SI for 2 yrs = $PR^2/100^2$ · doubling under SI at $R\%$ takes $100/R$ years
Average	Avg = sum/n · first n naturals = $(n+1)/2$ · weighted avg
Mixture/alligation	Cheaper:Dearer = $(\text{Mean}-C):(D-\text{Mean})$
Number system	Divisibility rules 3,4,8,9,11 · $\text{HCF} \times \text{LCM} = \text{product of two numbers}$ · sum of first n naturals = $n(n+1)/2$
P&C / Probability	$nPr = n!/(n-r)!$ · $nCr = n!/(r!(n-r)!)$ · $P = \text{favourable/total}$ · circular arrangement of $n = (n-1)!$

Reasoning tactics

- **Seating**: draw the diagram, place **definite** information first, then conditional. Circular facing centre → left is clockwise. **Hard 90-second limit per set** — then guess all and move on.
- **Syllogism**: use Venn diagrams. "Some A are B" → "Some B are A" is valid. "All A are B" does NOT give "All B are A". Possibility cases follow from the definite diagram.
- **Blood relations**: draw a family tree, use + for male and – for female.
- **Direction**: sketch it; remember the person's facing direction changes left/right.

English — Active to Passive

Tense	Active	Passive
Simple Present	writes	is/am/are + written
Present Continuous	is writing	is/am/are being written
Present Perfect	has written	has/have been written
Simple Past	wrote	was/were written
Past Continuous	was writing	was/were being written
Past Perfect	had written	had been written
Simple Future	will write	will be written
Future Perfect	will have written	will have been written
Modals	can/must write	can/must be written

No passive form exists for: future continuous, present perfect continuous, past perfect continuous, future perfect continuous. Intransitive verbs have no passive. Imperative → "Let + object + be + V3".

GK — Awards hierarchy

- Civilian: **Bharat Ratna > Padma Vibhushan > Padma Bhushan > Padma Shri**. Announced on **26 January**, conferred by the President. Padma awards are for **any field** — art, science, sport, medicine, social work, public affairs. Bharat Ratna: max 3/year, cannot be used as a title/prefix/suffix.
- Sports: **Major Dhyan Chand Khel Ratna** (highest, renamed 2021) · Arjuna (performance) · Dronacharya (coaches) · Dhyan Chand (lifetime).
- Military: **Param Vir Chakra** (highest wartime) > Maha Vir Chakra > Vir Chakra · **Ashoka Chakra** (highest peacetime).
- Others: Dadasaheb Phalke (cinema) · Jnanpith (literature) · Shanti Swarup Bhatnagar (science) · Gandhi Peace Prize · Ramon Magsaysay.

9. Exam-Day Tactics

1. **Read the instruction page first**. Confirm total questions, duration, and the marking scheme. Reports indicate 60 discipline + 40 aptitude with **no negative marking**, but the admit card is authoritative.
2. **If there is no negative marking, leave ZERO blanks**. Reserve the final 5 minutes purely to mark an option on every untouched question.
3. **Three passes**. Pass 1: answer everything you know instantly (~45 min). Pass 2: return to the workable ones. Pass 3: guess the rest.
4. **Abandon aggressively**. Any single question over 90 seconds gets marked and skipped. One seating set is worth the same as one BCNF question.
5. **Cutoff is only 50 / 45 / 40 marks**. You do not need to be perfect — you need breadth. Do not sink time chasing one hard topic.
6. **Start with your strongest section** to build momentum and bank easy marks early.

Carry: printed admit card, original photo ID, photographs, and anything else the admit card lists. Reach the centre early — verify the address and travel time today, not tomorrow morning.